SwaraSetu — Benchmark Report

deterministic WHO IMCI engine vs trained TF-IDF+LogReg · generated 2026-08-21 21:43 UTC · commit 68bf3b1

NOT CLINICALLY VALIDATED. Research prototype benchmarked on a synthetic/internal English case set and one public physician-labeled vignette corpus. Nothing here is evidence of clinical safety or effectiveness. Do not use for patient care.

1. Baseline performance (original pipeline, untouched)

Metric	Value
Accuracy	0.0675
Macro Precision	0.4108
Macro Recall	0.3539
Macro F1	0.0772
Weighted F1	0.0863
HIGH precision	0.6163
HIGH recall	0.0647
HIGH F1	0.1171
HIGH→LOW errors	732
HIGH→MEDIUM errors	34

Held-out set: triage_protocols_structured.json (2251 cases). Model: deterministic WHO IMCI engine (backend/app/triage/engine.py @ 68bf3b1)

2. Improved performance (experiment_improved v1)

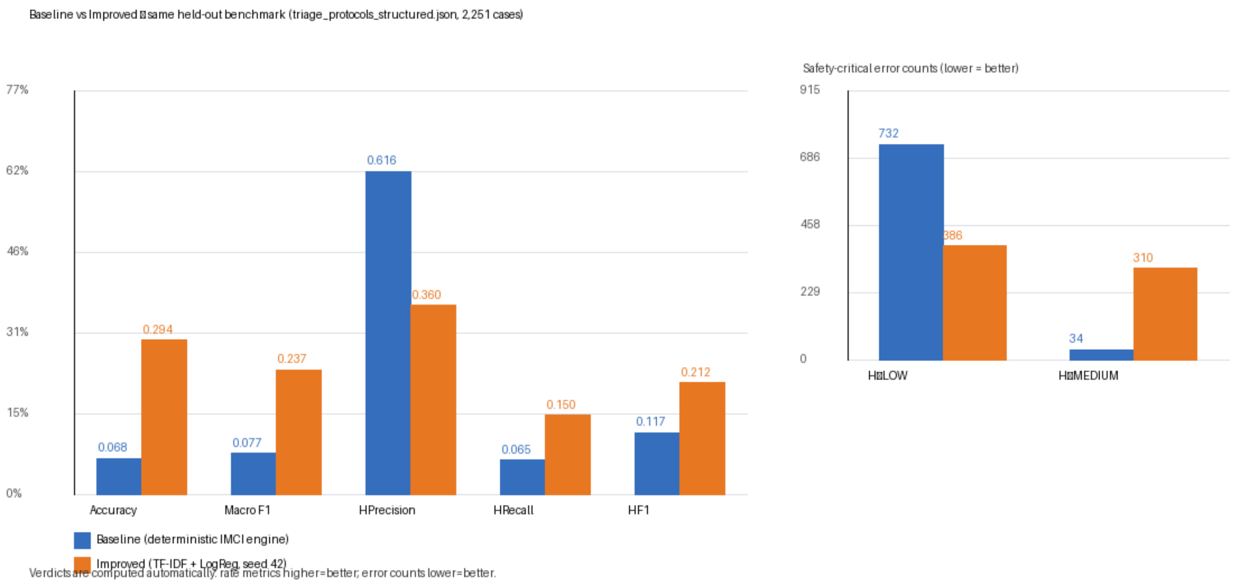
Metric	Value
Accuracy	0.2941
Macro Precision	0.3292
Macro Recall	0.2957
Macro F1	0.2369
Weighted F1	0.3640
HIGH precision	0.3596
HIGH recall	0.1502
HIGH F1	0.2119
HIGH→LOW errors	386
HIGH→MEDIUM errors	310

Held-out set: triage_protocols_structured.json (2251 cases). Model: experiment_improved_v1 (tfidf_logreg seed=42) — NOT the IMCI engine

3. Before vs After — identical held-out set

Metric	Baseline	Improved	Δ	Verdict
Accuracy	0.0675	0.2941	0.2266	**improved**
Macro F1	0.0772	0.2369	0.1597	**improved**
HIGH Precision	0.6163	0.3596	-0.2566	**regressed**
HIGH Recall	0.0647	0.1502	0.0855	**improved**
HIGH F1	0.1171	0.2119	0.0948	**improved**
HIGH -> LOW errors	732	386	-346	**improved**
HIGH -> MEDIUM errors	34	310	276	**regressed**

Error-count rows: lower is better (see notes column in CSV). Two honest regressions: HIGH precision ↓ and HIGH→MEDIUM errors ↑.



Verdicts are computed automatically: rate metrics higher=better; error counts lower=better.

NOT clinically validated. Improved model trained only on synthetic public data; see registry.

4. Confusion matrices (rows = ground truth, columns = predicted)

Baseline (IMCI engine)	pred HIGH	pred MED	pred LOW	Σ
true HIGH	53	34	732	819

Baseline (IMCI engine)	pred HIGH	pred MED	pred LOW	Σ
true MEDIUM	32	51	1299	1382
true LOW	1	1	48	50
Σ predicted	86	86	2079	2251

Improved (TF-IDF + LogReg)	pred HIGH	pred MED	pred LOW	Σ
true HIGH	123	310	386	819
true MEDIUM	209	521	652	1382
true LOW	10	22	18	50
Σ predicted	342	853	1056	2251

5 & 6. Safety-critical indicators (internal held-out benchmark)

Indicator	Baseline	Improved	Interpretation
HIGH-risk recall	6.47%	15.02%	both clinically inadequate (<20%)
HIGH→LOW errors	732	386	missed emergencies triaged to self-care
HIGH→MEDIUM errors	34	310	improved model shifts misses upward but still misses

7. Language performance (reports/language_benchmark.csv)

Subset	Eval type	Model	n	Acc	Macro F1	HI rec	H→L err
english	held_out	baseline_imci_engine	2251	0.0675	0.0772	0.0647	732
english	held_out	improved_tfidf_logreg	2251	0.2941	0.2369	0.1502	386
english	in_sample	improved_tfidf_logreg	4570	0.5127	0.5115	0.7254	185
english	unseen_for_imci	baseline_imci_engine*	4570	0.3707	0.2141	0.0000	1099
hindi	in_sample	improved_tfidf_logreg	4532	0.5181	0.5139	0.7759	162
hindi	unseen_for_imci	baseline_imci_engine*	4532	0.3588	0.2082	0.0000	1145
code_mixed	in_sample	improved_tfidf_logreg	4821	0.5177	0.5166	0.7320	161
code_mixed	unseen_for_imci	baseline_imci_engine*	4821	0.3634	0.2092	0.0000	1157
tamil	-	-	0	N/A	N/A	N/A	N/A
telugu	-	-	0	N/A	N/A	N/A	N/A
kannada	-	-	0	N/A	N/A	N/A	N/A
malayalam	-	-	0	N/A	N/A	N/A	N/A
marathi	-	-	0	N/A	N/A	N/A	N/A
bengali	-	-	0	N/A	N/A	N/A	N/A
transliteration	-	-	0	N/A	N/A	N/A	N/A

* baseline engine consumes raw text through the production rule-NER fallback. Rows marked in_sample were part of improved-model training (SYNTHETIC public data) — they are NOT validation results. N/A = no labelled data available; nothing fabricated.

8. External validation (unseen during training)

Dataset	Model	n	Acc(strict)	Acc(lenient)	Macro F1	HI rec	H→L	H→M
ai triage benchmark 78vign	improved tfidf logreg	78	0.3205	0.5000	0.2681	0.2000	11	21
ai triage benchmark 78vign	baseline imci engine via	78	0.1538	0.1795	0.1324	0.2000	30	2
physionet mietic	all	-	-	-	-	-	-	-
mimic iv ed	all	-	-	-	-	-	-	-
yale ed triage hong2018	all	-	-	-	-	-	-	-
nhamcs ed	all	-	-	-	-	-	-	-
displace m	all	-	-	-	-	-	-	-

Only ai_triage_benchmark_78vignettes (public GitHub; physician-labeled A–D scale, sha256 dd0d6aec…) qualifies as truly unseen external test data today. Mapping: A→LOW, B/C→MEDIUM, D→HIGH (documented; source labels preserved). Strict accuracy uses the most severe element of split labels; lenient credits any valid label. Other approved sets await credentialed/manual access.

9 & 10. Dataset sizes and sources (data/dataset_registry.yaml)

Dataset	Badge	Languages	Size	License	Train/Test/ExtTe
swarasetu triage protocols 225	INTERNAL-BENCHMARK		2251 cases	unspecified (provenance undocumented i	F/T/F
tulsiandhare multilingual medi	SYNTHETIC-PUBLIC	en, hi, hi-en (code-mixed)	13923 rows observed in loc	unspecified on HF card (synthetic data	T/T/F
physionet mietic	PUBLIC-CREDENTIALED		9629 cases	PhysioNet credentialed health data lic	T/T/T
mimic iv ed	PUBLIC-CREDENTIALED		~425k ED stays (v2.2)	PhysioNet credentialed health data lic	T/T/T
yale ed triage hong2018	PUBLIC-CREDENTIALED		560486 visits	no explicit OSS license; citation of H	F/T/T
nhamcs ed	PUBLIC-CREDENTIALED		~20k ED visits/yr; 1992-20	public-use; statistical reporting/anal	F/T/T
ihqid webmd 1mg	PUBLIC-CREDENTIALED	hi, bn, ta, te, gu, mr	WebMD + 1mg FAQ queries	academic release; explicit license not	F/F/F
indicmeddialog	PUBLIC-CREDENTIALED	as, bn, gu, hi, mr, pa,	2980 dialogues x 10 langua	not stated on landing page	F/F/F
himed freedomintelligence	PUBLIC-CREDENTIALED		corpus 403,516 + bench 7,7	not stated in README at audit time	F/F/F

Dataset	Badge	Languages	Size	License	Train/Test/ExtTe
coild health v2	PUBLIC	hi + "19 Indic targets"	unknown (not downloaded)	not stated on card at audit time	F/F/F
ekacare bodhi s	PUBLIC-CREDENTIALED	en	779 conditions / 4037 symp	CC BY-NC 4.0	T/T/F
ekacare indicmteb	PUBLIC-CREDENTIALED	en, hi, bn, ta, te, kn, mr,	2532 doctor-verified queri	CC BY-SA 4.0 (per Eka release blog)	T/T/F
displace m	PUBLIC-CREDENTIALED	hi, en (code-mixed Hinglish)	~35h dev+eval audio; 260 s	challenge dataset terms; HF-hosted for	T/T/T
ai triage benchmark 78vignette	PUBLIC-CREDENTIALED	en	78 cases (44 consensus + 3	no explicit OSS license observed in re	F/T/T
ai4bharat indicvoices	PUBLIC	22 scheduled Indic languages	23.7k hrs raw / 11.2k tran	CC BY 4.0 (AI4Bharat standard; confirm	T/T/F
ai4bharat kathbath	PUBLIC	12 Indic languages	1684 hrs, 1218 speakers, 2	CC BY 4.0 (AI4Bharat standard; confirm	T/T/F
local bengali banglish triage	QUARANTINED	bn (script), banglish (roman	2724 dialogues	UNKNOWN	F/F/F

Clinical reference material used to BUILD the engine: WHO IMCI protocol (published guidelines) — it is not a dataset and confers no validation. T/F flags: training/testing/external-test allowed per registry policy.

11. Limitations (read before citing any number above)

1. **No clinical validation.** No clinician reviewed outputs; IMCI implementation is untested against real patients.
2. Internal benchmark provenance is undocumented; legacy GT mapping maps BLACK (deceased) to LOW.
3. Baseline keyword adapter loses most signal — 91.74% under-triage internally; engine predicts LOW for ~92% of inputs.
4. Improved model trained ONLY on synthetic multilingual data; its Tulsiandhare numbers are in-sample and optimistic.
5. External validation covers 78 English vignettes; strict accuracy 32% (improved) / 15% (baseline via rule-NER) — far below deployment thresholds.
6. Language coverage gap: labelled triage data exists for en/hi/code-mixed only; ta/te/kn/ml/mr/bn/transliteration are N/A, not zero.
7. Frontend offline engine has drifted from backend (missing snake-bite/trauma rules added in 68bf3b1).
8. Known latent bug: extract_symptoms_rule_fallback can raise TypeError on pregnant+headache transcripts (invalid field name).
9. Approved credentialed corpora (MIETIC, MIMIC-IV-ED, Yale ETD, NHAMCS, DISPLACE-M) not yet ingested.
10. Edge ASR falls back to canned fixture transcripts without ONNX weights; latency figures exclude real decoding.

Reproduce: see README.md §Benchmarks — eval_baseline.py · train_improved.py · eval_improved.py · compare_before_after.py · eval_multilingual.py · run_external_validation.py · build_report_pdf.py